

GLOBAL CONFLICTS – IMPACT ON SUPPLY CHAINS

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Introduction

1. In this project we have studied how armed conflicts disrupt global shipping and their supply routes. Near some conflict zones, ships stop broadcasting their AIS signal or broadcast a false one, and become invisible to normal tracking. Using satellite radar vessel detections, we have identified which ships in five conflict-affected corridors (the Strait of Hormuz, Black Sea, Red Sea, Gulf of Aden and Arabian Sea) were visible to radar but not identifying themselves on AIS. We compared them against a worldwide baseline and linked the results to weekly conflict data from 2014 to 2026. To show the supply-chain impact, we added the Black Sea Grain Initiative, oil flows through key chokepoints and IMF trade data on Red Sea rerouting.

Key Terms

2. Here are the key terms frequently used in the report:

Term	Meaning in this project
AIS	A VHF transponder system that broadcasts a ship's identity (MMSI), position, course and speed. Under SOLAS Chapter V, Regulation 19, it is mandatory for ships of 300 gross tonnage and above on international voyages, cargo ships of 500 GT and above, and all passenger ships.
SAR detection	A vessel seen by a radar satellite (Sentinel-1). Radar detects the hull whether or not the ship is transmitting, so it acts as an independent "ground truth".
Unmatched detection (dark candidate)	A SAR detection for which no AIS position could be matched. The ship is physically present but not identifying itself on AIS.
AIS dead zone	An area where ships transmit but their signals are not received (coverage or reception gap). Expected signature: vessels of all sizes are unmatched at similar rates.
AIS dark spot	An area where ships that should be visible are not: AIS switched off, or positions falsified by spoofing or GPS jamming. Expected signature: large, AIS-mandated vessels are unmatched far more than normal.
Identification crisis	A situation in which a large share of the ships in an area cannot be identified through AIS, typically triggered by conflict.

Scope

3. Five conflict-relevant maritime corridors are analysed: Persian Gulf / Strait of Hormuz, Black Sea, Red Sea, Gulf of Aden / Bab-el-Mandeb, and Arabian Sea / western Indian Ocean. All SAR detections outside these five corridors form the rest-of-world baseline (labelled "Other / open water" in the dashboard), which defines what a "normal" unmatched rate looks like.

Methodology

4. **Core metric: the AIS-unmatched rate.** Global Fishing Watch matches each SAR detection to AIS positions broadcast near the same place and time. A detection with a matched MMSI is "AIS matched"; one without is "AIS unmatched". The core metric throughout the dashboard is the **AIS-unmatched rate**: unmatched detections divided by all detections in a group (a corridor, a size band, or a grid cell). Across all 107,257 detections, 28,889 (26.9%) are unmatched and 73.1% are matched. All counts in this report are SAR detections, not unique vessels: a ship imaged on several days is counted each time, so the number of distinct ships is lower.

5. **Corridor definition.** Each detection is assigned to a corridor by latitude/longitude bounding boxes, implemented as a DAX calculated column and replicated exactly in Python. Boxes are checked in the order below; a detection falling in two boxes takes the first match.

Order	Corridor	Latitude	Longitude	Detections
1	Persian Gulf / Strait of Hormuz	22° to 30.5°N	47° to 58°E	5,814
2	Red Sea	12° to 30°N	32° to 44°E	1,033
3	Gulf of Aden / Bab-el-Mandeb	9° to 15°N	43° to 52°E	209
4	Black Sea	40° to 47.5°N	27° to 42°E	3,499
5	Arabian Sea / Indian Ocean (west)	0° to 25°N	52° to 78°E	3,431
–	Rest-of-world baseline ("Other / open water")	all other	all other	93,271

6. **Controlling for vessel size.** A raw unmatched rate mixes two very different groups: small boats, many of which are not required to carry AIS, and large ships, which must. To separate them, every detection is placed in a **size band** by SAR-estimated length: Small (<25 m), Medium (25–50 m), Large (50–100 m) and Very large (100 m+). Length of 50 m or more is used as a practical proxy for AIS-mandated ships, and SAR detects vessels of this size very reliably (Paolo et al., Nature, 2024, report detection rates above 90% for vessels of 50 m and larger). The headline comparisons use 100m+ ships, where an unmatched detection is hardest to explain innocently.

7. **Separating dead zones from dark spots.** The size × corridor matrix (Page 3) applies three reading rules. If a corridor is **unmatched at high rates across all sizes**, the likely cause is a reception gap: an AIS dead zone. If only **small boats** are unmatched, the cause is non-AIS craft, which is normal. If **large ships (50m+)** are unmatched far above the baseline, the cause is deliberate switching-off or falsified positions: a dark spot and an identification crisis.

8. **Linking visibility to conflict.** Conflict countries are linked to corridors through a bridge table that maps each conflict country to the corridor it borders (Iran → Hormuz; Yemen → Red Sea and Gulf of Aden; Ukraine → Black Sea; “Indian Ocean” → Arabian Sea). For each corridor we compute an **escalation sharpness** measure in DAX: the busiest week of conflict events divided by the average week for the mapped country. A value of 12× means the worst week had twelve times the normal weekly number of events. This is compared with the large-ship (50m+) unmatched rate in a scatter plot. Because there are only five corridors, the relationship is summarised with Spearman's rank correlation ($\rho = 0.82$, $p = 0.09$) and treated as suggestive, not as proof. The peak week is taken over the full 2014–2026 period, so this measure describes how volatile each conflict is, not its intensity during the SAR window.

9. **Statistical testing.** For 100m+ ships in each corridor, the unmatched rate is reported with a 95% Wilson score confidence interval, which stays accurate for small samples and for rates near 0% or 100%. Each corridor is compared with the rest-of-world baseline using Fisher's exact test, which is exact rather than approximate and therefore valid even for the Gulf of Aden (130 large ships). Because five corridors are tested, p-values are adjusted with a Bonferroni correction (multiplied by 5) so that "significant" results are not found by chance. An overall chi-square test of independence checks whether large-ship visibility differs across corridors at all ($\chi^2 = 5,042.4$, 5 degrees of freedom, $p < 0.001$).

10. **Predictive model.**

10.1. The model answers one question for every ship of 50 m or longer (58,594 detections): is this ship AIS-dark? It is given seven features: the great-circle (haversine) distance to five chokepoints (Strait of Hormuz, Kerch Strait, Bab-el-Mandeb, Suez Canal, Bosphorus), the ship's length, and the local traffic density (log of detections in the same 0.5° cell). Two algorithms are compared: logistic regression, a simple baseline that can only learn straight-line relationships, and histogram-based gradient boosting, which combines hundreds of small decision trees and can learn curved relationships such as "very dark up to 500 km, then normal".

10.2. The models are evaluated with spatial block cross-validation. The map is divided into $5^\circ \times 5^\circ$ blocks and, in each of five folds, the model is tested only on blocks it did not see during training (scikit-learn GroupKFold). A random split would let the model memorise neighbouring ships and overstate accuracy; the spatial split tests whether the pattern generalises to new places. Performance is measured by the area under the ROC curve (AUC), where 0.5 is random guessing and 1.0 is perfect. Permutation importance measures how much the model worsens when each feature is shuffled, showing which factors drive darkness.

10.3. Two outputs come from this stage. The distance-decay analysis measures the observed unmatched rate of 50m+ ships in distance rings (0–100 km, 100–200 km, and so on up to 1,000 km) around Hormuz, Kerch and Bab-el-Mandeb, giving the radius of each dark zone. The risk grid asks the final model, for each 0.5° cell in the region ($0-48^\circ\text{N}$, $25-80^\circ\text{E}$), how likely a 150 m ship there would be dark, and labels each cell Low ($<15\%$), Elevated ($15-30\%$), High ($30-50\%$) or Very high ($>50\%$).

10.4. Supply-chain context and route mapping. Supply-chain impact is shown through three sources used for their own periods: the Black Sea Grain Initiative (what flowed and who depended on it), EIA oil-chokepoint flows (what is at stake at each chokepoint) and IMF PortWatch (how trade rerouted during the Red Sea crisis). Shipping routes are drawn from a Power Query M script that routes each path through real chokepoints (Bosphorus, Dardanelles, Suez, Bab-el-Mandeb, Malacca, Hormuz) and interpolates points every 0.8° so the paths render as continuous lines. These routes are schematic: they show which chokepoints each flow depends on, not actual vessel tracks.

Dashboard Walkthrough

11. The dashboard has six pages that build one argument in sequence: the conflict landscape (Page 1), what "normal" fishing activity looks like (Page 2), where ships are invisible and which ones (Page 3), the Black Sea precedent and what is at stake (Page 4), how routes shifted and the verdict for each corridor (Page 5), and a model that predicts where dark zones form (Page 6).

11.1. Page 1 – Conflict Supply Chain Command

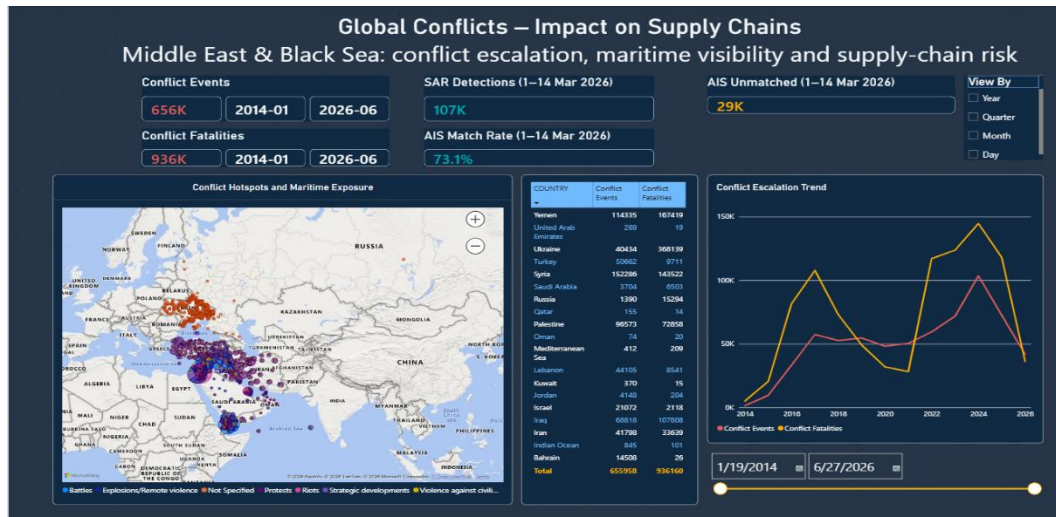


Figure 1. Page 1 – Global Conflicts: Impact on Supply Chains

Purpose. The entry page. It sets out the conflict landscape across the Middle East and the Black Sea and places the headline maritime-visibility numbers beside it.

11.1.1. Conflict KPI cards (selected period)

What it shows. Total conflict events and fatalities for the period chosen on the date slider, with two small cards showing the start and end month of the selection.

What it tells us. Gives the scale of conflict behind the maritime analysis, and lets any sub-period (for example, the February 2022 invasion of Ukraine) be examined on its own.

11.1.2. SAR KPI cards (1–14 Mar 2026)

What it shows. SAR Detections (107,257), AIS Match Rate (73.1%) and AIS Unmatched detections (28,889).

How to read it. These cards are fixed to the SAR window and are deliberately not affected by the date slider; this was tested.

What it tells us. Roughly one in four ships seen by radar in the window could not be identified through AIS. The following pages explain where, and which ships.

11.1.3. View By slicer

What it shows. A field parameter that switches the time axis of the trend chart between Year, Quarter, Month and Day.

11.1.4. Map – Conflict Hotspots and Maritime Exposure

What it shows. Conflict events plotted at admin-1 region centroids, coloured by event type (battles, explosions/remote violence, protests, riots, strategic developments, violence against civilians) and sized by number of events. "Not specified" marks the Ukraine–Russia source, which does not carry event types.

How to read it. Larger bubbles mean more events. Look at how close the clusters sit to the sea.

What it tells us. Several of the most intense clusters border the maritime corridors: Yemen overlooks Bab-el-Mandeb, Iran borders the Strait of Hormuz, and Ukraine and Crimea border the Black Sea.

11.1.5. Country table

What it shows. Conflict events and fatalities by country for the selected period.

How to read it. Sorted by events; fatalities show intensity.

What it tells us. Over the full period Syria has the most events (152,286), but the countries that matter for shipping are Yemen (114,335 events, 167,419 fatalities), Iran (41,798 events) and Ukraine, which has the highest fatality count in the data (368,139).

11.1.6. Conflict Escalation Trend

What it shows. Conflict events (red) and fatalities (yellow) over time, at the level chosen in View By.

How to read it. The 2026 value covers January to June only, so a lower 2026 total does not mean de-escalation.

What it tells us. At year level, events peak in 2024 (about 100,000) and fatalities peak in 2017 and 2024. At month level, the Russian invasion of Ukraine is visible as a fatality spike in February 2022, and conflict events reach their highest monthly level of 2025–26 in March 2026 – the same month as the SAR window. The radar snapshot therefore captures shipping during an active escalation.

11.2. Page 2 – Fishing Fleet Baseline (2020)

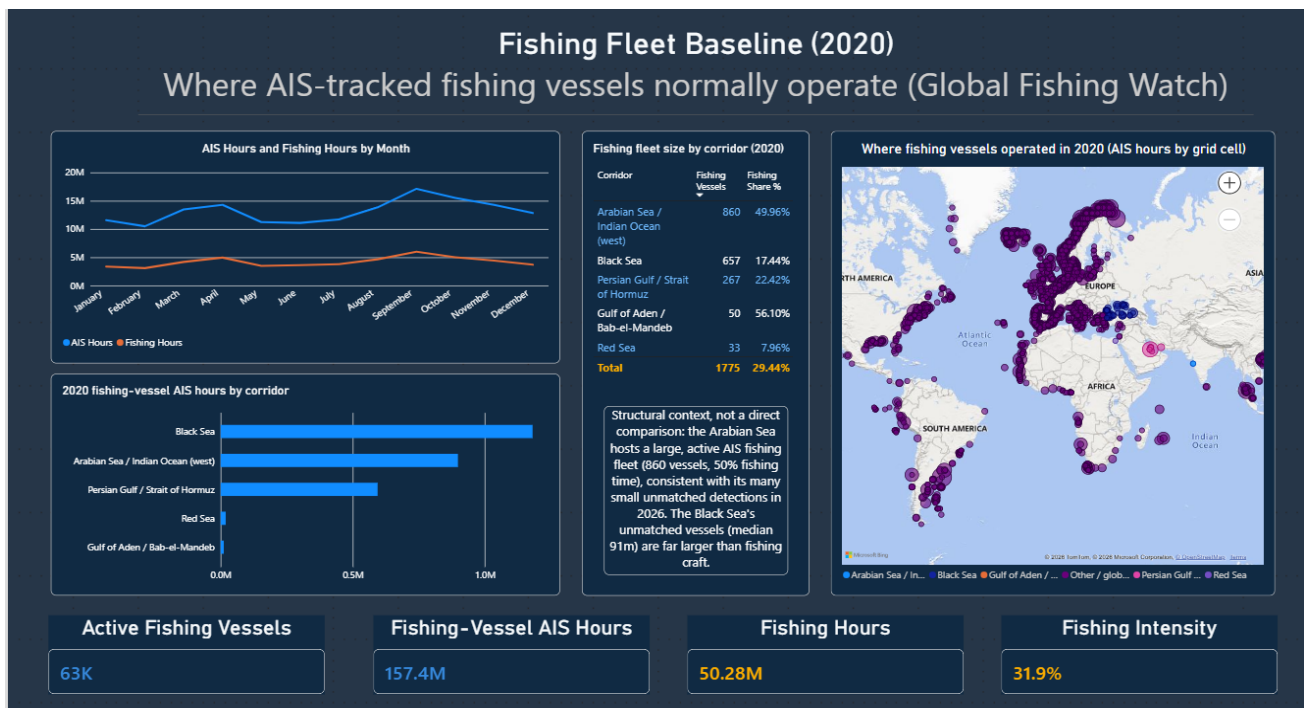


Figure 2. Page 2 – Fishing Fleet Baseline (2020)

Purpose. Shows where AIS-tracked fishing vessels normally operate. This helps explain the many small unmatched detections in some corridors. It is structural context only and is never joined to the 2026 data.

11.2.1. AIS Hours and Fishing Hours by Month

What it shows. Total AIS hours and fishing hours of fishing vessels each month of 2020.

How to read it. The gap between the two lines is time at sea spent not fishing (transit, searching).

What it tells us. Activity is fairly steady (11–17 million AIS hours per month) with a peak in September. Seasonality is small, so a two-week window in March is not an unusual time of year.

11.2.2. 2020 fishing-vessel AIS hours by corridor

What it shows. AIS hours of fishing vessels inside each of the five corridors.

How to read it. Longer bars mean more AIS-equipped fishing activity.

What it tells us. The Black Sea (about 1.2 million hours) and the Arabian Sea (about 0.9 million) have large fishing fleets; the Red Sea and Gulf of Aden have almost none on AIS.

11.2.3. Fishing fleet size by corridor (2020)

What it shows. Number of distinct fishing vessels (by MMSI) and fishing share (fishing hours ÷ AIS hours) per corridor.

What it tells us. The Arabian Sea has 860 fishing vessels fishing about 50% of their time at sea – a large, active fleet. The Black Sea has 657 vessels

(17% fishing time), the Persian Gulf 267, the Gulf of Aden 50 and the Red Sea 33.

11.2.4. Map – Where fishing vessels operated in 2020

What it shows. AIS hours by 0.1° grid cell, coloured by corridor.

What it tells us. Fishing is concentrated along continental shelves and coasts, with the densest activity in Europe, East Asia and South America.

11.2.5. KPI cards

What it shows. Active fishing vessels (63K), fishing-vessel AIS hours (157.4 million), fishing hours (50.28 million) and fishing intensity (31.9% of AIS time spent fishing).

Key insight (text box on the page). The Arabian Sea's large, active fishing fleet is consistent with its many small unmatched detections in 2026. The Black Sea's unmatched vessels (median length 91 m) are far too large to be fishing craft, so its darkness has a different cause.

11.3. Page 3 – Conflict-AIS Visibility Watch

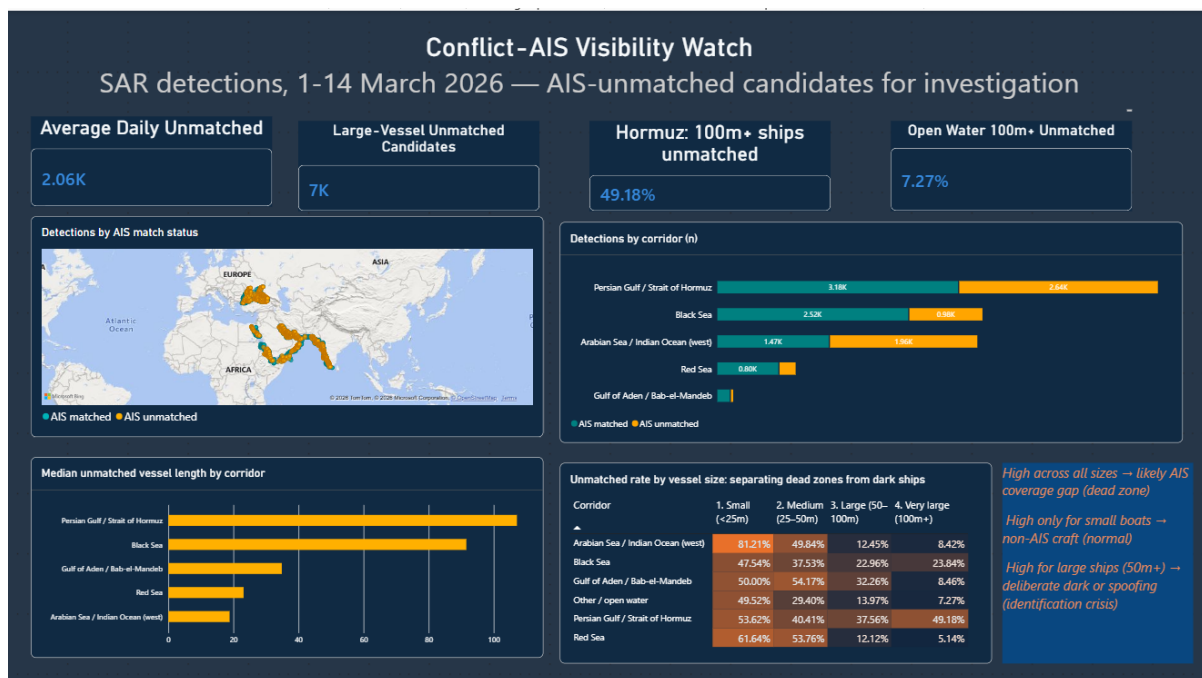


Figure 3. Page 3 – Conflict-AIS Visibility Watch

Purpose. The core analytical page. It shows where ships were invisible to AIS during 1–14 March 2026, how many, and – most importantly – which ones.

11.3.1. KPI cards

What it shows. Average daily unmatched detections (2,063), large-vessel unmatched candidates (7,374 unmatched detections of vessels 50 m or longer), the Hormuz 100m+ unmatched rate (49.18%) and the rest-of-world 100m+ unmatched rate (7.27%).

How to read it. The last two cards are placed side by side on purpose: they are the headline contrast of the project.

What it tells us. In normal waters about 1 in 14 large ships is unidentified; near Hormuz it is about 1 in 2. The 7,374 large unmatched detections form a ready-made watchlist for investigation. Daily counts are not analysed as a trend because they mostly reflect which areas the satellite imaged each day.

11.3.2. Map – Detections by AIS match status

What it shows. Every SAR detection in the five corridors, teal if AIS-matched and orange if unmatched.

What it tells us. Unmatched detections concentrate in the Persian Gulf, along the Arabian Sea coasts and in the northern Black Sea.

11.3.3. Detections by corridor (n)

What it shows. Number of detections per corridor, split into matched and unmatched.

How to read it. The bar length is the sample size; the orange share is the unmatched part.

What it tells us. Hormuz has the most detections (5,814, of which 2,638 unmatched), followed by the Black Sea (3,499) and the Arabian Sea (3,431). The Gulf of Aden has only 209 detections, so its rates must be read with caution.

11.3.4. Median unmatched vessel length by corridor

What it shows. The typical (median) length of the unmatched ships in each corridor.

How to read it. A long median means the unmatched ships are large, AIS-mandated vessels; a short one means small boats.

What it tells us. Unmatched ships near Hormuz have a median length of about 107 m and in the Black Sea about 91 m – commercial-scale vessels. In the Arabian Sea the median is about 19 m: small craft.

11.3.5. Unmatched rate by vessel size (heatmap matrix)

What it shows. The unmatched rate for every combination of corridor and size band, coloured from dark navy (low) to orange (high).

How to read it. Apply the three rules printed beside it: high across all sizes means a dead zone; high only for small boats is normal; high for large ships means deliberate darkness or spoofing.

Corridor	Small <25m	Medium 25–50m	Large 50–100m	Very large 100m+
Persian Gulf / Strait of Hormuz	53.6%	40.4%	37.6%	49.2%
Black Sea	47.5%	37.5%	23.0%	23.8%
Gulf of Aden / Bab-el-Mandeb	50.0%	54.2%	32.3%	8.5%

Corridor	Small <25m	Medium 25–50m	Large 50–100m	Very large 100m+
Arabian Sea / Indian Ocean (west)	81.2%	49.8%	12.5%	8.4%
Red Sea	61.6%	53.8%	12.1%	5.1%
Rest-of-world baseline	49.5%	29.4%	14.0%	7.3%

Table 1. AIS-unmatched rate by corridor and vessel size, 1–14 March 2026

What it tells us. Everywhere else, the bigger the ship, the more visible it is – as expected, because large ships must carry AIS. Hormuz reverses this pattern: its largest ships are the least visible (49.2%, about 7× the baseline). The Black Sea is the only other corridor where large ships are clearly elevated (about 3×). The Arabian Sea shows the classic small-craft pattern (81% of small boats unmatched, but only 8% of 100m+ ships). No corridor is high across all sizes, so no classic dead zone was found. External reporting supports the timing: Windward recorded more than 1,100 ships affected by GPS jamming in the Gulf within 24 hours of the 28 February 2026 escalation, rising to about 1,650 by 7 March – inside our SAR window [10].

11.4. Black Sea Precedent & Corridor Risk

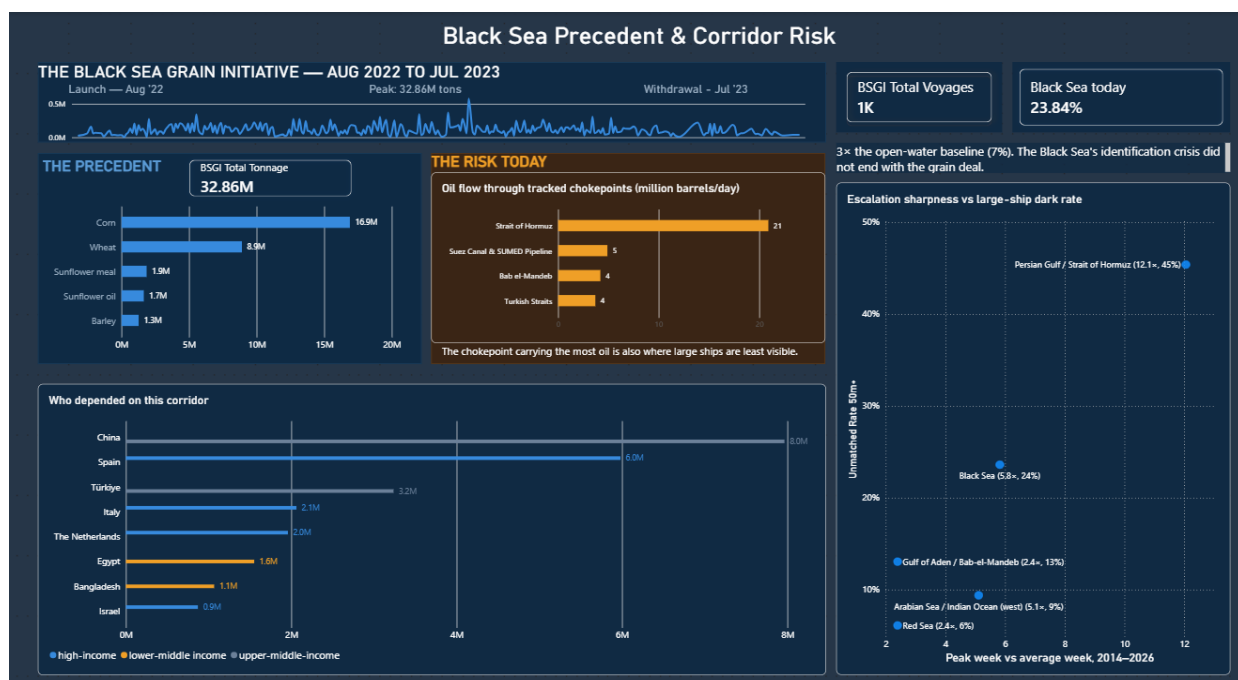


Figure 4. Page 4 – Black Sea Precedent & Corridor Risk

Purpose. Uses the Black Sea Grain Initiative as a documented precedent of a conflict-disrupted corridor, shows that the Black Sea's identification problem continues today, and connects conflict escalation to large-ship darkness across all five corridors.

11.4.1 BSGI timeline (Aug 2022 – Jul 2023)

What it shows. Daily grain tonnage shipped under the Black Sea Grain Initiative, from launch in August 2022 to Russia's withdrawal in July 2023.

What it tells us. Shows how a negotiated corridor can keep trade moving through an active war zone – and how quickly it can stop.

11.4.2. The Precedent – tonnage, voyages and commodities

What it shows. Total tonnage (32.86 million tonnes), total voyages (about 1,000) and tonnage by commodity.

What it tells us. Corn (16.9 million t, about 51%) and wheat (8.9 million t, about 27%) dominate, followed by sunflower meal, sunflower oil and barley.

11.4.3. Who depended on this corridor

What it shows. BSGI tonnage by destination country, coloured by World Bank income group.

What it tells us. China (8.0 million t) and Spain (6.0 million t) were the largest buyers, but lower-middle-income importers such as Egypt (1.6 million t) and Bangladesh (1.1 million t) show the food-security exposure of poorer countries to a Black Sea disruption.

11.4.4. The Risk Today – Oil flow through tracked chokepoints

What it shows. Oil flow in million barrels per day through the chokepoints linked to the five corridors (EIA).

What it tells us. The Strait of Hormuz carries about 21 million barrels per day – far more than Suez & SUMED (5), Bab-el-Mandeb (4) or the Turkish Straits (4). The chokepoint carrying the most oil is also where large ships are least visible.

11.4.5. Black Sea today card

What it shows. 23.84% of 100m+ ships in the Black Sea were unmatched in March 2026.

What it tells us. About three times the rest-of-world baseline (7.3%). The Black Sea's identification problem did not end with the grain deal.

11.4.6. Escalation sharpness vs large-ship dark rate (scatter)

What it shows. Each dot is a corridor. The horizontal axis is escalation sharpness (peak conflict week ÷ average week, 2014–2026); the vertical axis is the unmatched rate of 50m+ ships. Labels give both values.

How to read it. Dots further right had sharper conflict spikes; dots higher up had more large ships going dark.

What it tells us. Hormuz (12.1×, 45%) and the Black Sea (5.8×, 24%) are highest on both axes; the Red Sea (2.4×, 6%) and Gulf of Aden (2.4×, 13%) are lowest. The rank correlation is strong (Spearman $\rho = 0.82$) but, with only five corridors, not statistically significant ($p = 0.09$). The exception is the Arabian Sea (5.1×, 9%): a real escalation spike, yet large ships stayed visible. That exception points to the more precise pattern in Key Findings: darkness follows the type of actor (state militaries with jamming capability), not escalation alone.

11.5. Before, During, Now: Where the Ships Went

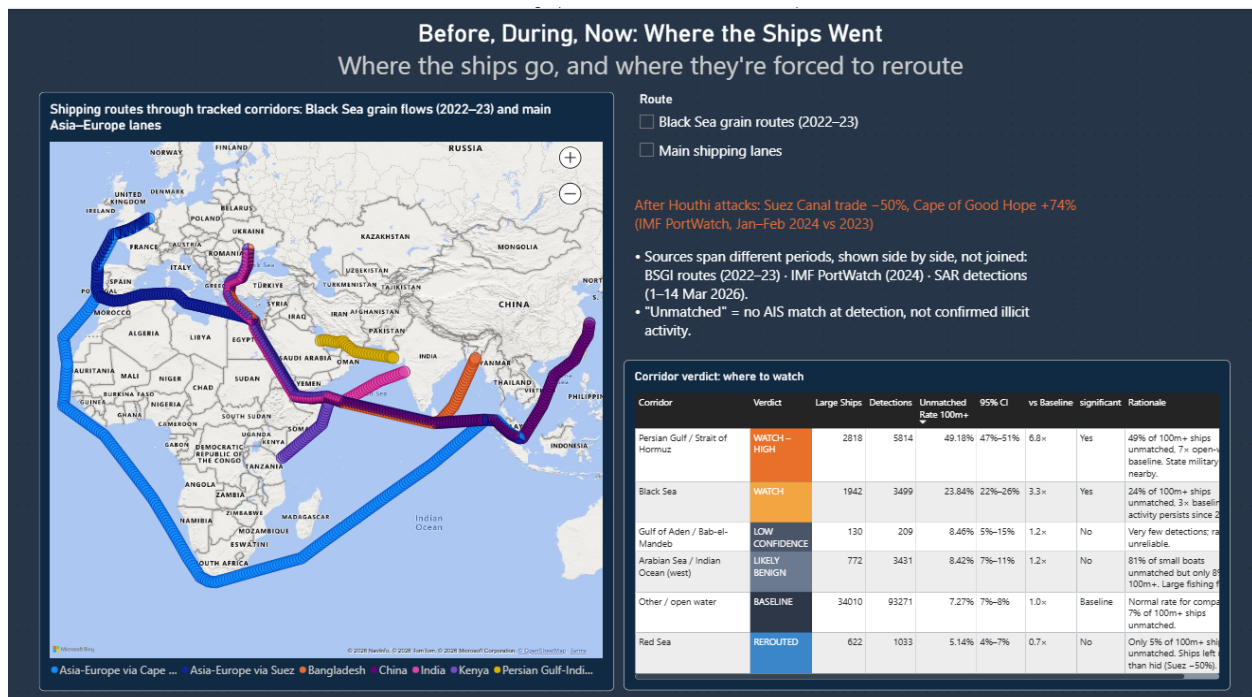


Figure 5. Page 5 – Before, During, Now: Where the Ships Went

. Tells the rerouting story in three steps – where ships used to go, what changed, and which corridors need watching now – and delivers the final verdict for each corridor.

11.5.1. Route map with Route slicer

What it shows. Schematic shipping routes through the tracked corridors. "Black Sea grain routes (2022–23)" shows BSGI flows to China, India, Bangladesh and Kenya; "Main shipping lanes" shows Asia–Europe via Suez, Asia–Europe via the Cape of Good Hope, and the Persian Gulf–India oil route. The slicer switches between the two layers; with nothing selected, both are shown.

How to read it. Every eastbound grain route shares one trunk through the Bosphorus, Suez, the Red Sea and Bab-el-Mandeb before splitting. The Cape route shows the much longer alternative.

What it tells us. Before: grain and container trade to Asia depended on a single chokepoint chain. During: after the Houthi attacks, ships switched to the Cape route (see IMF callout).

11.5.2. IMF PortWatch callout

What it shows. Suez Canal trade fell 50% and Cape of Good Hope trade rose 74% (January–February 2024 vs the same months of 2023) [5].

What it tells us. Provides the evidence that rerouting actually happened; the map alone shows only that the alternative route exists.

11.5.3. Corridor verdict: where to watch

What it shows. For each corridor: verdict, number of large (100m+) ships, total detections, 100m+ unmatched rate, 95% confidence interval, multiple of the baseline, statistical significance, and rationale. Verdict cells are colour-coded.

How to read it. Sorted by the 100m+ unmatched rate. "Statistically significant" means the corridor differs from the baseline after Bonferroni correction.

Corridor	Verdict	100m+ dark rate (95% CI)	vs baseline	Significant?
Persian Gulf / Strait of Hormuz	WATCH – HIGH	49.2% (47–51%)	6.8×	Yes
Black Sea	WATCH	23.8% (22–26%)	3.3×	Yes
Gulf of Aden / Bab-el-Mandeb	LOW CONFIDENCE	8.5% (5–15%)	1.2×	No
Arabian Sea / Indian Ocean (west)	LIKELY BENIGN	8.4% (7–11%)	1.2×	No
Rest-of-world	BASELINE	7.3% (7–8%)	1.0×	–
Red Sea	REROUTED	5.1% (4–7%)	0.7×	No

Table 2. Corridor verdicts with statistical evidence

What it tells us. Only two corridors are genuinely abnormal: for Hormuz and the Black Sea, even the lower end of the confidence interval is far above the baseline's upper end. The Red Sea's large ships are at or below normal visibility – the ships still there are not hiding, the rest have left. The Arabian Sea is benign for large ships. The Gulf of Aden's interval is wide (5–15%) because only 130 large ships were detected, hence "low confidence". A note on the page explains that sources from different periods are shown side by side but not joined, and that "unmatched" means no AIS match at detection, not confirmed illicit activity.

11.6. Predicting AIS Dark Zones

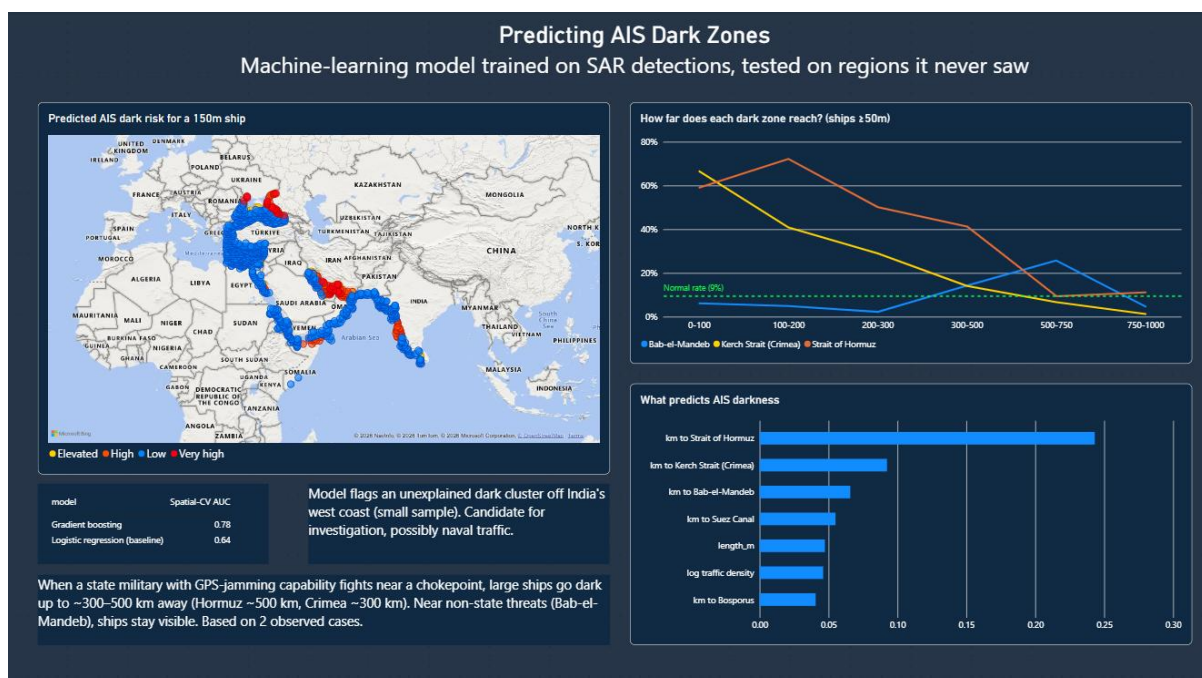


Figure 6. Page 6 – Predicting AIS Dark Zones

Purpose. Moves from describing where ships were dark to predicting where they will be, using the machine-learning model and distance analysis from Python (see Python Analysis).

11.6.1. Map – Predicted AIS dark risk for a 150m ship

What it shows. The model's predicted probability that a 150 m ship in each 0.5° cell would be AIS-dark, grouped into Low, Elevated, High and Very high risk (896 cells: 685 Low, 66 Elevated, 108 High, 37 Very high).

How to read it. Red and orange cells are where a large, AIS-mandated ship is most likely to be unidentified.

What it tells us. High and very-high risk cells cluster around the Persian Gulf and Strait of Hormuz and in the northern Black Sea around Crimea. A small, unexpected cluster also appears off India's west coast (see below).

11.6.2. How far does each dark zone reach? (ships ≥50m)

What it shows. The observed unmatched rate of 50m+ ships in distance rings around three chokepoints, with a constant line at the normal rate (about 9%).

How to read it. Where a line falls to the normal-rate line marks the edge of that dark zone.

Distance from chokepoint	Strait of Hormuz	Kerch Strait (Crimea)	Bab-el-Mandeb
0–100 km	59%	67%	6%
100–200 km	72%	41%	5%
200–300 km	50%	29%	2%
300–500 km	41%	14%	14%
500–750 km	9%	7%	26% (74 ships)
750–1,000 km	11%	1%	5%

Table 3. Unmatched rate of 50m+ ships by distance from each chokepoint

What it tells us. The Hormuz dark zone reaches about 500 km, beyond which rates fall back to normal. The Crimea dark zone reaches about 300 km. Around Bab-el-Mandeb, large ships are at or below the normal rate, so there is no dark zone at all; the 500–750 km ring is based on only 74 ships and overlaps other waters.

11.6.3. Model accuracy on unseen regions (AUC)

What it shows. Spatial cross-validation AUC for the two models: gradient boosting 0.78, logistic regression 0.64.

What it tells us. Gradient boosting clearly outperforms the straight-line baseline. The dark-zone effect is non-linear (strong up to a radius, then normal), which a linear model cannot capture. A score of 0.78 on regions the model never saw shows that the pattern generalises.

11.6.4. What predicts AIS darkness

What it shows. Permutation importance of each feature.

What it tells us. Distance to the Strait of Hormuz is by far the strongest predictor (0.243), followed by distance to the Kerch Strait (0.092) and Bab-el-Mandeb

(0.066). Ship length and traffic density matter much less. Where a ship is, relative to a state-level conflict chokepoint, matters more than what kind of ship it is.

Prediction statement (text box on the page). When a state military with GPS-jamming capability fights near a chokepoint, large ships go dark up to roughly 300–500 km away (Hormuz ~500 km, Crimea ~300 km). Near non-state threats such as Bab-el-Mandeb, ships stay visible and reroute instead. This statement is based on two observed cases.

Lead for investigation (text box on the page). The risk map flags an unexplained cluster of dark large ships off India's west coast: 37% of 19 large ships at 12–14°N and 93% of 14 large ships at 16–18°N, while the busier Mumbai–Gujarat waters (18–24°N, 454 large ships) are at 2–5%. There is no conflict here. One possible explanation is naval traffic, which is not required to broadcast AIS; the Karwar naval base lies on this coast at about 14.8°N. The sample is small, so this is presented as a lead to investigate, not a conclusion.

Python Analysis

12. Two Python scripts extend the dashboard with statistical testing and machine learning. They were run in VS Code (Python 3); the full code is in Appendices A and B. Results are saved as CSV files and loaded into Power BI (Pages 5 and 6).

12.1. Libraries Used

- 12.1.1. Pandas
- 12.1.2. NumPy
- 12.1.3. SciPy
- 12.1.4. Statsmodels
- 12.1.5. Scikit-learn
- 12.1.6. Pathlib

12.2. Scripts Written

12.2.1. **Statistical Testing (step19_stats.py).** The script loads the raw SAR file, checks the date window and missing values, assigns corridors with the same boxes and order as the DAX column, flags unmatched detections (no MMSI) and assigns size bands. It then prints a validation check against Power BI and, for 100m+ ships, computes each corridor's unmatched rate, Wilson interval, Fisher's exact test against the baseline, Bonferroni-adjusted p-value and the overall chi-square test. Outputs: **corridor_stats.csv** (loaded into the verdict table) and **sar_clean.csv** (input to Script 2).

Table 4. Script 1 output – chi-square test: $\chi^2 = 5,042.4$, $df = 5$, $p < 0.001$

Corridor	n (100m+)	Unmatched	Rate	95% CI	× baseline	p (Bonferroni)
Persian Gulf / Strait of Hormuz	2,818	1,386	49.2%	47.3–51.0%	6.77	< 0.001
Black Sea	1,942	463	23.8%	22.0–25.8%	3.28	< 0.001
Gulf of Aden / Bab-el-Mandeb	130	11	8.5%	4.8–14.5%	1.17	1.00

Corridor	n (100m+)	Unmatched	Rate	95% CI	× baseline	p (Bonferroni)
Arabian Sea / Indian Ocean (west)	772	65	8.4%	6.7–10.6%	1.16	1.00
Rest-of-world baseline	34,010	2,470	7.3%	7.0–7.5%	1.00	–
Red Sea	622	32	5.1%	3.7–7.2%	0.71	0.21

12.2.2. Predictive model (step20_model.py). The script keeps ships of 50 m or more (58,594 detections, 12.6% unmatched), computes the seven features, runs five-fold spatial block cross-validation for both models, fits the final gradient-boosting model, calculates permutation importance, measures distance decay around three chokepoints and scores the regional risk grid. Outputs: model_metrics.csv, feature_importance.csv, distance_decay.csv and risk_grid.csv (all used on Page 6).

Model	Spatial-CV AUC	Interpretation
Logistic regression (baseline)	0.64	Weak: a straight-line model cannot describe a dark zone with a radius
Histogram gradient boosting	0.78	Good: generalises to 5° blocks it never saw during training

Table 5. Model comparison

Feature	Permutation importance
Distance to Strait of Hormuz	0.243
Distance to Kerch Strait (Crimea)	0.092
Distance to Bab-el-Mandeb	0.066
Distance to Suez Canal	0.055
Ship length (m)	0.047
Log traffic density	0.046
Distance to Bosphorus	0.041

Table 6. Feature importance (drop in AUC when the feature is shuffled)

Predicted rates for held-out regions closely track the observed rates: for example, the Black Sea is observed at 23.6% (50m+) and predicted at 28.5%, the Red Sea observed at 6.1% and predicted at 5.2%, and the baseline observed at 9.4% and predicted at 8.8%.

12.3. Integration with Power BI. Each CSV is imported through Get Data → Text/CSV. corridor_stats is related one-to-one to Corridor_Verdict on the corridor name, and two DAX columns format the interval and the baseline multiple for display. The risk grid is plotted on a map using its latitude and longitude, and the distance-decay table is sorted by a band-order column so that distance rings appear in the correct order.

Key Findings

13. The key findings from the dashboard and the Python analysis, with the evidence that supports each one, are:

13.1. The Strait of Hormuz shows an identification crisis: its largest ships are its least visible. (49.2% of 100m+ unmatched, 6.8× baseline, $p < 0.001$ (P3, P5)).

13.2. The Black Sea is the second hotspot, and the problem has outlasted the grain deal. (23.8% of 100m+ unmatched, 3.3×, $p < 0.001$; dark zone within ~300 km of Kerch (P4, P6))

13.3. No classic dead zone exists in the five corridors; the excess darkness is behavioural or electronic. (No corridor high across all size bands (P3))

13.4. The Red Sea was rerouted rather than darkened. (5.1% of 100m+ unmatched; Suez -50%, Cape +74% (P5))

13.5. The Arabian Sea's high raw unmatched rate (57%) is explained by small fishing craft. (81% of <25m unmatched but 8% of 100m+; 860-vessel fishing fleet (P2, P3))

13.6. Darkness follows the type of actor, not escalation alone. (Dark hotspots near Iran and Russia; none near the Houthis despite escalation (P4, P6))

13.7. Dark zones have a measurable radius and can be predicted. (Hormuz ~500 km, Crimea ~300 km; AUC 0.78 on unseen regions (P6))

13.8. The most oil flows exactly where visibility is lowest. (Hormuz ~21 million barrels/day (P4))

Defence Implications

14. Beyond the statistical findings, the analysis has several practical implications for maritime security and for the Indian Armed Forces:

14.1. **AIS cannot be trusted near state-level conflict chokepoints.** Within roughly 300–500 km of Hormuz and Crimea, between a quarter and three-quarters of large ships are not identifiable on AIS. Maritime domain awareness in these zones should treat AIS as incomplete and cross-check it with satellite radar, patrol aircraft and other sensors. The 7,374 large unmatched detections in a single fortnight show the scale of the task.

14.2. **A rising large-ship dark rate is an early-warning indicator.**

Because darkness appears within the radius of a state military with electronic-warfare capability, a sudden rise in the share of unmatched large ships near a chokepoint can signal the start of jamming or spoofing – and therefore of an escalation – before it is reported.

14.3. **The threat type determines the shipping response.** Near a state-level threat, ships stay on the route but become unidentifiable, creating a surveillance problem. Near a non-state threat, ships stay identifiable but leave the route, creating a cost and capacity problem (longer Cape voyages). Planners should prepare for both.

14.4. **Energy and food security exposure.** The highest-risk zone sits on the chokepoint carrying about 21 million barrels of oil per day, including the Persian Gulf–India oil route. The BSGI shows that lower-middle-income importers such as Egypt and Bangladesh are exposed when a grain corridor is disrupted.

14.5. **A lead in India's own waters.** The model flags an unexplained dark cluster off India's west coast. Checking it against naval and coast-guard movement records would test whether the model can also highlight sensitive activity outside conflict zones.

Bibliography

15. Following are the different sources from which datasets have been referred to and used in the project.
 - 15.1. [Datathon 2026 – Conflicts and Supply Chains](https://cdm.gov.in/index.php/2026/08/06/datathon-2026/) (<https://cdm.gov.in/index.php/2026/08/06/datathon-2026/>) – source of the ACLED Middle East aggregated conflict data
 - 15.2. HarborIQ: Maritime Traffic Intelligence (https://www.kaggle.com/datasets/kavyachauhan05/harboriq-maritime-traffic-intelligence?select=ships_cleaned.csv) – reviewed, not used (no timestamps, invalid identity fields)
 - 15.3. mmsi-daily-csvs-10-v2-2020 (<https://www.kaggle.com/datasets/artemsmirnov1109/mmsidailycsvs10v22020>)
 - 15.4. indian-ocean-sar-vessel-detections (<https://www.kaggle.com/datasets/shubh5106/indian-ocean-sar-vessel-detections>)
 - 15.5. Red Sea crisis: Suez Canal traffic impacts (<https://data.humdata.org/dataset/the-red-sea-suez-canal-traffic>)
 - 15.6. India: Daily Port Activity Data and Shipment Estimates (<https://data.humdata.org/dataset/india-daily-port-activity-data-and-shipment-estimates>) – reviewed, not used in this stage
 - 15.7. UCDP Georeferenced Event Dataset (GED) Global (<https://ucdp.uu.se/downloads/#section-yearly-1989>)
 - 15.8. U.S. Energy Information Administration – World Oil Transit Chokepoints (https://www.eia.gov/international/analysis/special-topics/World_Oil_Transit_Chokepoints)
 - 15.9. United Nations – Black Sea Grain Initiative, Joint Coordination Centre data (<https://www.un.org/en/black-sea-grain-initiative/data>)
 - 15.10. Woodward (2026) – GPS Jamming Disrupts 1,100 Ships in the Middle East Gulf (<https://woodward.ai/blog/gps-jamming-disrupts-1100-ships-in-the-middle-east-gulf/>) and GPS Jamming Surges in the Middle East Gulf, 1,650 Ships Hit (<https://woodward.ai/blog/gps-jamming-surges-in-the-middle-east-gulf-1650-ships-hit/>)
 - 15.11. Software: Microsoft Power BI Desktop; Python 3 with pandas, NumPy, SciPy, statsmodels and scikit-learn; Measure Killer (Brunner BI).

1 Appendix A – Python Script 1: step19_stats.py

Statistical testing of 100m+ unmatched rates by corridor. Change the two paths at the top to your own folders.

```
import pandas as pd
import numpy as np
from pathlib import Path
from scipy.stats import fisher_exact, chi2_contingency
from statsmodels.stats.proportion import proportion_confint

# ---- SETTINGS ----
CSV_PATH = r"D:\Datathon\sar_vessel_detections_pipev4_202603.csv"
OUT_DIR = Path(r"D:\Datathon\python_outputs")
OUT_DIR.mkdir(exist_ok=True)

# ---- LOAD + CLEAN ----
df = pd.read_csv(CSV_PATH)
raw_rows = len(df)
df["timestamp"] = pd.to_datetime(df["timestamp"], utc=True, errors="coerce")
start, end = pd.Timestamp("2026-03-01", tz="UTC"), pd.Timestamp("2026-03-15", tz="UTC")
in_window = df["timestamp"].between(start, end, inclusive="left")
removed_date = int((~in_window).sum())
df = df[in_window]
missing_xy = int(df[["lat", "lon"]].isna().any(axis=1).sum())
df = df.dropna(subset=["lat", "lon"])
missing_len = int(df["length_m"].isna().sum())

# ---- SAME CORRIDORS AS POWER BI (same order as the DAX SWITCH) ----
lat, lon = df["lat"], df["lon"]
conds = [
    lat.between(22, 30.5) & lon.between(47, 58),
    lat.between(12, 30) & lon.between(32, 44),
    lat.between(9, 15) & lon.between(43, 52),
    lat.between(40, 47.5) & lon.between(27, 42),
    lat.between(0, 25) & lon.between(52, 78),
]
names = ["Persian Gulf / Strait of Hormuz", "Red Sea", "Gulf of Aden / Bab-el-Mandeb",
         "Black Sea", "Arabian Sea / Indian Ocean (west)"]
df["Corridor"] = np.select(conds, names, default="Other / open water")
df["unmatched"] = df["mmsi"].isna()
df["size_band"] = pd.cut(df["length_m"], bins=[0, 25, 50, 100, np.inf], right=False,
                        labels=["Small (<25m)", "Medium (25-50m)", "Large (50-100m)", "Very
large (100m+)"])

# ---- CLEANING LOG ----
print("=== CLEANING LOG ===")
print(f"Raw rows in CSV: {raw_rows:,}")
print(f"Removed (outside 1-14 Mar 2026): {removed_date:,}")
print(f"Removed (missing lat/lon): {missing_xy:,}")
print(f"Rows analysed: {len(df):,}")
print(f"Missing length (kept, excluded from size analysis): {missing_len:,}")

# ---- CHECK AGAINST POWER BI ----
print("\n=== CHECK AGAINST POWER BI ===")
```

```

print(f"Overall unmatched rate: {df['unmatched'].mean():.1%} (Power BI: ~26.9%)")
h = df[(df["Corridor"] == names[0]) & (df["length_m"] >= 100)]
print(f"Hormuz 100m+ unmatched: {h['unmatched'].mean():.2%} (Power BI: 49.18%)")

# ---- STATISTICS: 100m+ ships, each corridor vs baseline ----
big = df[df["length_m"] >= 100]
base = big[big["Corridor"] == "Other / open water"]
k0, n0 = int(base["unmatched"].sum()), len(base)
rate0 = k0 / n0

rows = []
for corr, g in big.groupby("Corridor"):
    k, n = int(g["unmatched"].sum()), len(g)
    lo, hi = proportion_confint(k, n, alpha=0.05, method="wilson")
    p = np.nan if corr == "Other / open water" else fisher_exact([[k, n - k], [k0, n0 - k0]])[1]
    rows.append({"Corridor": corr, "n_100m": n, "unmatched_100m": k,
                "dark_rate": k / n, "ci_low": lo, "ci_high": hi,
                "x_baseline": (k / n) / rate0, "p_value": p})

stats = pd.DataFrame(rows)
n_tests = stats["p_value"].notna().sum()
stats["p_bonferroni"] = (stats["p_value"] * n_tests).clip(upper=1)
stats["significant"] = np.select(
    [stats["p_value"].isna(), stats["p_bonferroni"] < 0.05], ["Baseline", "Yes"], default="No")
stats = stats.sort_values("dark_rate", ascending=False)

ct = pd.crosstab(big["Corridor"], big["unmatched"])
chi2, p_chi, dof, _ = chi2_contingency(ct)

print("\n=== 100m+ DARK RATE BY CORRIDOR (95% Wilson CI, Fisher exact vs
baseline) ===")
print(stats.round(4).to_string(index=False))
print(f"\nChi-square test (corridor vs AIS match, 100m+ ships): chi2={chi2:.1f}, dof={dof},
p={p_chi:.2e}")

# ---- SAVE FOR POWER BI + NEXT STEP ----
stats.to_csv(OUT_DIR / "corridor_stats.csv", index=False)
df[["lat", "lon", "length_m", "Corridor", "size_band", "unmatched"]].to_csv(OUT_DIR /
"sar_clean.csv", index=False)
print(f"\nSaved: {OUT_DIR / 'corridor_stats.csv'} and sar_clean.csv")

```

2 Appendix B – Python Script 2: step20_model.py

Spatial machine-learning model, distance decay and risk grid. Input is sar_clean.csv produced by Script 1.

```
import pandas as pd
import numpy as np
from pathlib import Path
from sklearn.model_selection import GroupKFold
from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import HistGradientBoostingClassifier
from sklearn.preprocessing import StandardScaler
from sklearn.pipeline import make_pipeline
from sklearn.metrics import roc_auc_score
from sklearn.inspection import permutation_importance

# ---- SETTINGS ----
CSV_PATH = r"D:\Datathon\python_outputs\sar_clean.csv" # output of script 1
OUT_DIR = Path(r"D:\Datathon\python_outputs")
OUT_DIR.mkdir(exist_ok=True)

# ---- DATA: ships >= 50m (AIS is mandatory for them) ----
df = pd.read_csv(CSV_PATH).dropna(subset=["length_m"])
df = df[df["length_m"] >= 50].copy()
df["unmatched"] = df["unmatched"].astype(bool)

# ---- FEATURES: distance to conflict chokepoints + traffic density + size ----
CHOKEPOINTS = {
    "Strait of Hormuz": (26.6, 56.3),
    "Kerch Strait (Crimea)": (45.3, 36.5),
    "Bab-el-Mandeb": (12.6, 43.4),
    "Suez Canal": (29.9, 32.6),
    "Bosporus": (41.1, 29.0),
}

def km(lat, lon, point):
    la, lo = np.radians(lat), np.radians(lon)
    pa, po = np.radians(point[0]), np.radians(point[1])
    a = np.sin((pa - la) / 2) ** 2 + np.cos(la) * np.cos(pa) * np.sin((po - lo) / 2) ** 2
    return 6371 * 2 * np.arcsin(np.sqrt(a))

feat_cols = []
for name, pt in CHOKEPOINTS.items():
    col = "km to " + name
    df[col] = km(df["lat"], df["lon"], pt)
    feat_cols.append(col)

df["cell"] = (np.floor(df["lat"] / 0.5) * 0.5).round(1).astype(str) + "_" + (np.floor(df["lon"] / 0.5) * 0.5).round(1).astype(str)
df["log traffic density"] = np.log1p(df.groupby("cell")["lat"].transform("size"))
feat_cols += ["length_m", "log traffic density"]

X, y = df[feat_cols].values, df["unmatched"].values
# spatial blocks of 5 degrees: the model is always tested on regions it never saw
blocks = (np.floor(df["lat"] / 5)).astype(int).astype(str) + "_" + (np.floor(df["lon"] / 5)).astype(int).astype(str)
```

```

# ---- MODELS: spatial block cross-validation ----
models = {
    "Logistic regression (baseline)": make_pipeline(StandardScaler(),
    LogisticRegression(max_iter=1000)),
    "Gradient boosting": HistGradientBoostingClassifier(max_iter=300,
    learning_rate=0.05, random_state=0),
}
metrics = []
for name, model in models.items():
    oof = np.zeros(len(y))
    for tr, te in GroupKFold(n_splits=5).split(X, y, blocks):
        model.fit(X[tr], y[tr])
        oof[te] = model.predict_proba(X[te])[0, 1]
    auc = roc_auc_score(y, oof)
    metrics.append({"model": name, "spatial_cv_auc": round(auc, 3)})
    print(f"{name}: spatial-CV AUC = {auc:.3f}")
    df["pred_" + name] = oof

# ---- FINAL MODEL + FEATURE IMPORTANCE ----
final = HistGradientBoostingClassifier(max_iter=300, learning_rate=0.05,
random_state=0).fit(X, y)
pi = permutation_importance(final, X, y, n_repeats=5, random_state=0,
scoring="roc_auc")
importance = pd.DataFrame({"feature": feat_cols, "importance":
pi.importances_mean}).sort_values("importance", ascending=False)
print("\nWhat drives AIS darkness (permutation importance):")
print(importance.round(3).to_string(index=False))

print("\nObserved vs predicted (held-out) dark rate by corridor:")
print(df.groupby("Corridor")[["unmatched", "pred_Gradient boosting"]].mean().round(3))

# ---- DISTANCE DECAY: how far does each dark zone reach? ----
bands = [0, 100, 200, 300, 500, 750, 1000, 1500, 2000, 20000]
labels = ["0-100", "100-200", "200-300", "300-500", "500-750", "750-1000", "1000-1500",
"1500-2000", "2000+"]
rows = []
for name in ["Strait of Hormuz", "Kerch Strait (Crimea)", "Bab-el-Mandeb"]:
    b = pd.cut(df["km to " + name], bands, labels=labels)
    g = df.groupby(b, observed=True)[["unmatched"]].agg(["size", "mean"]).reset_index()
    for i, r in g.iterrows():
        rows.append({"chokepoint": name, "distance_km": r.iloc[0], "band_order":
labels.index(r.iloc[0]),
                    "n_ships": int(r["size"]), "dark_rate": round(r["mean"], 4)})
decay = pd.DataFrame(rows)
decay = decay[decay["band_order"] <= 5] # keep 0-1000 km: beyond that, other
hotspots mix in
print("\nDark rate by distance (ships >= 50m):")
print(decay.pivot(index="distance_km", columns="chokepoint",
values="dark_rate").reindex(labels).round(3))

# ---- RISK GRID: predicted dark probability for a 150m ship in each 0.5 deg cell ----
region = df[df["lat"].between(0, 48) & df["lon"].between(25, 80)]
cells = region.groupby("cell").agg(lat=("lat", "mean"), lon=("lon", "mean"),
n_detections=("lat", "size"),
observed_dark_rate=("unmatched", "mean"),

```

```

log_density=("log traffic density", "first").reset_index()
grid = pd.DataFrame({c: km(cells["lat"], cells["lon"], pt) for c, pt in
                      {"km to " + n: p for n, p in CHOKEPOINTS.items()}.items()})
grid["length_m"] = 150
grid["log traffic density"] = cells["log_density"]
cells["predicted_dark_risk"] = final.predict_proba(grid[feat_cols].values)[: , 1].round(4)
cells["risk_band"] = pd.cut(cells["predicted_dark_risk"], [0, 0.15, 0.3, 0.5, 1],
                             labels=["Low", "Elevated", "High", "Very high"], include_lowest=True)

# ---- SAVE FOR POWER BI ----
pd.DataFrame(metrics).to_csv(OUT_DIR / "model_metrics.csv", index=False)
importance.to_csv(OUT_DIR / "feature_importance.csv", index=False)
decay.to_csv(OUT_DIR / "distance_decay.csv", index=False)
cells.drop(columns="log_density").to_csv(OUT_DIR / "risk_grid.csv", index=False)
print(f"\nSaved 4 files to {OUT_DIR}")
print(cells["risk_band"].value_counts())

```

Spearman rank correlation for the escalation scatter (Page 4) can be reproduced with:

```

from scipy.stats import spearmanr
# escalation sharpness (x) and 50m+ unmatched rate (y) for the five corridors
x = [12.1, 5.8, 5.1, 2.4, 2.4] # Hormuz, Black Sea, Arabian Sea, Gulf of Aden, Red Sea
y = [0.45, 0.24, 0.09, 0.13, 0.06]
print(spearmanr(x, y)) # rho = 0.82, p = 0.09

```

3 Appendix C – Key DAX Formulas

[Unmatched Rate] is the base measure: unmatched detections divided by all detections.

```
-- Calculated column: corridor assignment (SAR_202603)
Corridor =
SWITCH ( TRUE (),
    'SAR_202603'[lat] >= 22 && 'SAR_202603'[lat] <= 30.5 && 'SAR_202603'[lon] >= 47
    && 'SAR_202603'[lon] <= 58, "Persian Gulf / Strait of Hormuz",
    'SAR_202603'[lat] >= 12 && 'SAR_202603'[lat] <= 30 && 'SAR_202603'[lon] >= 32
    && 'SAR_202603'[lon] <= 44, "Red Sea",
    'SAR_202603'[lat] >= 9 && 'SAR_202603'[lat] <= 15 && 'SAR_202603'[lon] >= 43 &&
    'SAR_202603'[lon] <= 52, "Gulf of Aden / Bab-el-Mandeb",
    'SAR_202603'[lat] >= 40 && 'SAR_202603'[lat] <= 47.5 && 'SAR_202603'[lon] >= 27
    && 'SAR_202603'[lon] <= 42, "Black Sea",
    'SAR_202603'[lat] >= 0 && 'SAR_202603'[lat] <= 25 && 'SAR_202603'[lon] >= 52 &&
    'SAR_202603'[lon] <= 78, "Arabian Sea / Indian Ocean (west)",
    "Other / open water" )

-- Calculated column: vessel size band (SAR_202603)
Size Band =
SWITCH ( TRUE (),
    ISBLANK ( SAR_202603[length_m] ), "Unknown",
    SAR_202603[length_m] < 25, "1. Small (<25m)",
    SAR_202603[length_m] < 50, "2. Medium (25-50m)",
    SAR_202603[length_m] < 100, "3. Large (50-100m)",
    "4. Very large (100m+)" )

-- Measures: size-controlled AIS-unmatched rates
Detections = COUNTROWS ( SAR_202603 )
Unmatched Rate 50m+ = CALCULATE ( [Unmatched Rate], SAR_202603[length_m] >=
50 )
Unmatched Rate 100m+ = CALCULATE ( [Unmatched Rate], SAR_202603[Size Band]
= "4. Very large (100m+)" )
Hormuz 100m+ Unmatched = CALCULATE ( [Unmatched Rate], SAR_202603[Size
Band] = "4. Very large (100m+)", SAR_202603[Corridor] = "Persian Gulf / Strait of
Hormuz" )
Open Water 100m+ Unmatched = CALCULATE ( [Unmatched Rate], SAR_202603[Size
Band] = "4. Very large (100m+)", SAR_202603[Corridor] = "Other / open water" )
Black Sea 100m+ Unmatched = CALCULATE ( [Unmatched Rate], SAR_202603[Size
Band] = "4. Very large (100m+)", SAR_202603[Corridor] = "Black Sea" )

-- Measure: escalation sharpness (Conflict_Events)
Peak to Avg Ratio =
VAR WeeklySums = SUMMARIZE ( Conflict_Events, Conflict_Events[WEEK],
    "WeekEvents", CALCULATE ( SUM ( Conflict_Events[EVENTS] ) ) )
VAR AvgWeekly = AVERAGEX ( WeeklySums, [WeekEvents] )
VAR MaxWeekly = MAXX ( WeeklySums, [WeekEvents] )
RETURN DIVIDE ( MaxWeekly, AvgWeekly )

-- Calculated column: year-month axis (Conflict_Events)
Year Month = FORMAT ( Conflict_Events[WEEK], "YYYY-MM" )

-- Measures: 2020 fishing baseline (AIS_2020)
Fishing Vessels = DISTINCTCOUNT ( AIS_2020[mmsi] )
```

```
Fishing Share % = DIVIDE ( SUM ( AIS_2020[fishing_hours] ), SUM ( AIS_2020[hours] )
)
```

```
-- Calculated column: scatter labels (Corridor_Conflict_Bridge)
```

```
Corridor Label = Corridor_Conflict_Bridge[Corridor] & " (" & FORMAT ( [Peak to Avg  
Ratio], "0.0" ) & "x, " & FORMAT ( [Unmatched Rate 50m+], "0%" ) & ")"
```

```
-- Calculated columns: statistics display (corridor_stats, from Python)
```

```
95% CI = FORMAT ( corridor_stats[ci_low], "0%" ) & "-" & FORMAT ( corridor_stats[ci_high], "0%" )
```

```
vs Baseline = FORMAT ( corridor_stats[x_baseline], "0.0" ) & "x"
```